MGSC 2301 · PROFESSOR DENISE SPENCER · FRIDAY, OCTOBER 30, 2026

Business Statistics Exam 2

Chapters 6–8: the normal distribution, sampling distributions, and confidence intervals. Every formula is written exactly as it appears on the Canvas *Exam 2: Chapters 6-8 Formulas* sheet.

Exam format and rules

Source: Professor Spencer's “Exam 2 Preparation” announcement, Oct 28, 2026 · Daily Schedule · Formula Sheet for Test 2

COVERAGE	TIME	QUESTIONS
Chapters 6, 7, 8	95 minutes, in class. Closes automatically 5 min before class ends	50 questions, 59 pts, plus 3 extra credit at 1 pt each

QUESTION TYPES	ALLOWED	SOFTWARE
Q1–12 multiple choice. Q13–50 fill-in-the-blank	z-table, t-table, formula sheet (no notes), calculator, blank paper	Respondus LockDown Browser. Run the tech check first

- **Show handwritten work** on every fill-in question except the 13 that are marked exempt. Number each problem, put your name on every page, and hand in your work before you leave. Questions that require work get no credit without it.
- Enter every answer in Canvas. Your written work should match what you enter.
- Stay on the exam page. Leaving it can result in a zero. Work alone.
- Once you are in LockDown Browser, turn your phone off and put it in your bag. You can't use it as a calculator.
- The six exhibits contain 6, 13, 3, 6, 2, and 4 questions. Use each scenario only for its own grouped questions.
- Professor Spencer's tip: if you overthink multiple choice, start with the show-your-work questions, take the familiar points first, and come back to the confusing items.

The Canvas assignment list shows 66 possible points. That differs from the announcement, and this guide follows the professor's latest description.

The official formula sheet

Source: Formula Sheet for Test 2 (3 pages), reproduced in the same order, symbols, and labels

This is the sheet you get during the exam. Learn to recognize each line on sight. The grey note under each label says when to use it.

Exam 2: Chapters 6-8 Formulas

SHEET PAGE 1

$\sigma = \sqrt{\dfrac{\sum_{i=1}^{N} x_i^2 - \dfrac{(\Sigma x_i)^2}{n}}{N}} = \sqrt{\dfrac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N}}$	Standard Deviation Population. Divide by N.
$s = \sqrt{\dfrac{\sum_{i=1}^{n} x_i^2 - \dfrac{(\Sigma x_i)^2}{n}}{n-1}} = \sqrt{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$	Standard Deviation Sample. Divide by n − 1.
$\dfrac{\alpha}{2} \quad \& \quad 1 - \dfrac{\alpha}{2}$	Critical probability Left-tail and cumulative areas for the cutoffs of a CI, with α = 1 − confidence.
$z = \dfrac{x - \mu}{\sigma}$	*z*-score (raw data) One individual value (Chapter 6).
$\mathbf{x} = \mu + (\mathbf{z})(\sigma)$	*z*-score to a raw score Percentile or cutoff questions: find x.
$\mu = \mathbf{x} - (\mathbf{z})(\sigma)$	*z*-score to a population mean The mean is the unknown.
$E(\bar{x}) = \mu_{\bar{X}} = \mu$	Expected value for sampling distribution of $\bar{x}$ Center of the sample-mean distribution.

SHEET PAGE 2

$z_{\bar{x}} = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$	*z*-score (sample mean) Probability about an average of n items.
$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$	Standard error of the mean (σ known)
$\mu = \bar{x} \pm (Z_{\alpha/2}) \left(\frac{\sigma}{\sqrt{n}} \right)$	CI estimate for μ (σ known) Use z when the population σ is given.
$S_{\bar{X}} = \frac{s}{\sqrt{n}}$	Standard error of the mean (σ unknown)
$\mu = \bar{x} \pm (t_{a/2}) \left(\frac{s}{\sqrt{n}} \right)$	CI estimate for μ (σ unknown) Use t with df = n − 1. The sheet prints "a/2", which means α/2.
$\mathrm{n} = \left(\frac{(z_{\alpha/2})\sigma}{E} \right)^2$	Sample size for interval estimate of μ Always round UP.

SHEET PAGE 3

$\bar{p} = \frac{x}{n}$	Sample proportion
$E(\bar{p}) = p.$	Expected value for sampling distribution of $\bar{p}$
$\mathrm{np} \geq 5, \text{ and } \mathrm{n}(1 - \mathrm{p}) \geq 5$	Normality requirements for sampling distribution of $\bar{p}$
$z_{\bar{p}} = \frac{\bar{p} - p}{\sigma_{\bar{p}}}$	*z*-score (sample proportion)
$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$	Standard error of the proportion Uses the known population p.

$p = \bar{p} \pm (Z_{\alpha/2})\sqrt{\frac{\bar{p}(1-\bar{p})}{\mathrm{n}}}$	CI estimate for *p* p is unknown, so the SE uses $\bar{p}$.
$n = \frac{(z_{\alpha/2})^2 p^*(1-p^*)}{E^2}$	Sample size for interval estimate of *p* p* = pilot estimate, or .50 if none is given. Round UP.

The population line on the printed sheet shows $\bar{x}$ and n. For a population you are working with μ and N. Your exam copy will look exactly like the lines above.

Pick the right formula

Source: Formula sheet; Chapter 6–8 slides

First decide what the question is about: **one value, an average, or a proportion**. Then decide what it wants: **a probability, an interval, or a sample size**.

THE PROBLEM GIVES OR ASKS…	USE
Probability for one individual value x	$z = \frac{x-\mu}{\sigma}$
Cutoff value x from a percentile	Find z from the LEFT area, then $\mathrm{x} = \mu + (\mathrm{z})(\sigma)$
Unknown mean from x and a percentile	$\mu = \mathrm{x} - (\mathrm{z})(\sigma)$
Probability about a sample mean $\bar{x}$	$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$, then $z_{\bar{x}} = \frac{\bar{x}-\mu}{\sigma_{\bar{x}}}$
Probability about a sample proportion $\bar{p}$	$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$, then $z_{\bar{p}} = \frac{\bar{p}-p}{\sigma_{\bar{p}}}$
CI for a mean, σ given	$\mu = \bar{x} \pm (Z_{\alpha/2})\left(\frac{\sigma}{\sqrt{n}}\right)$
CI for a mean, only s given	$\mu = \bar{x} \pm (t_{a/2})\left(\frac{s}{\sqrt{n}}\right)$, df = n − 1
CI for a population proportion	$p = \bar{p} \pm (Z_{\alpha/2})\sqrt{\frac{\bar{p}(1-\bar{p})}{\mathrm{n}}}$
Minimum n to estimate a mean	$\mathrm{n} = \left(\frac{(z_{\alpha/2})\sigma}{E}\right)^2$, round UP

THE PROBLEM GIVES OR ASKS…	USE
Minimum n to estimate a proportion	$n = \frac{(z_{\alpha/2})^2 p^*(1-p^*)}{E^2}$, round UP

Three decisions that prevent most mistakes

1. **One value or an average?** A sample mean varies by $\sigma/\sqrt{n}$, not by σ. A story that mentions 50 people may still ask about just one person, so identify the actual random variable.
2. **σ or s?** For a mean CI, use t whenever the population σ is unknown, even if n > 30. Practice Test 2 uses t at n = 60 and n = 32. A large sample does not turn s into σ.
3. **Known p or estimated $\bar{p}$?** In a Chapter 7 probability, use the given population p in $\sigma_{\bar{p}}$. In a Chapter 8 CI, p is unknown, so use $\bar{p}$. For sample-size planning, use p* from a pilot study, or .50 when there isn't one.

Show-your-work template

Question number → list the givens → name the method and check normality → write the formula → substitute → show the SE, the z or t value, and E → give the final answer with the requested decimals and units. For a probability, sketch and shade the area. For an interval, label both endpoints and interpret it in terms of the population parameter.

Chapter 6 · Normal curves: values → areas

Source: DS Chapter 6 Normal Curve slides 3–24 · Chapter 6 PDF pp. 1–19 · Practice Test 2 goldfish problem

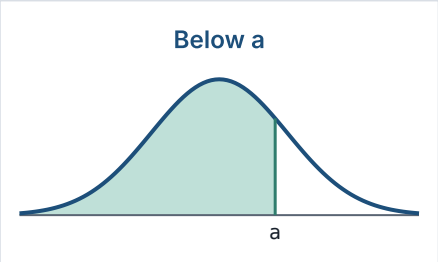
Facts to know without calculating

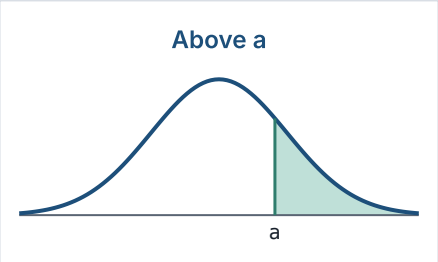
- A normal distribution is continuous, bell-shaped, symmetric, and described by μ and σ. Mean = median = mode.
- The total area is 1, and each side of the mean has area .50. The standard normal distribution has mean 0 and standard deviation 1.
- A positive z is above the mean and a negative z is below it.
- For a continuous variable, P(X = a) = 0, so P(X < a) = P(X ≤ a).
- Empirical rule: about 68%, 95%, and 99.7% of values fall within 1, 2, and 3σ of the mean. For an exact central 95%, use 1.96 rather than 2.

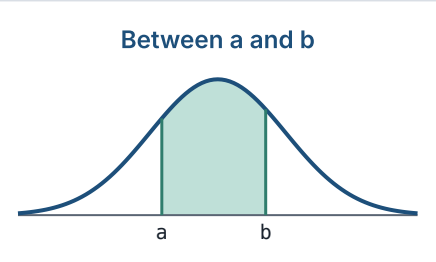
The course z-table gives the area to the LEFT

Compute $z = \dfrac{x - \mu}{\sigma}$ and round to two decimals. The row gives the ones and tenths digits, and the column gives the hundredths. *Example:* row 1.2, column .04 → .8925 for z = 1.24.

ASKED FOR	USE THE CUMULATIVE TABLE THIS WAY
Less than / below a	LEFT area at z(a)
Greater than / above a	1 − LEFT area at z(a)
Between a and b	LEFT area at z(b) − LEFT area at z(a)
Outside a to b	1 − (probability between a and b)







Course example: goldfish lifetimes, μ = 60 days, σ = 20 days

- **More than 60 days:** z = (60 − 60)/20 = 0, so P = **.50**.
- **More than 80 days:** z = (80 − 60)/20 = 1.00. LEFT area = .8413, so the right area = 1 − .8413 = **.1587** (15.87%).
- **Between 25 and 45 days:** z(25) = −1.75 → .0401, and z(45) = −.75 → .2266. Then .2266 − .0401 = **.1865**.
- **Replaced within 7 days:** z = (7 − 60)/20 = −2.65 → **.0040**, which is **.40%**. A probability and a percentage differ by a factor of 100.

> **Sanity checks.** A tail beyond the mean must be less than .50, and a far tail must be small. Every probability is between 0 and 1. To find an area between two values, subtract the *areas*, never the z-scores.

Chapter 6 · Percentiles: areas → values

Source: Chapter 6 slide 24 · Chapter 6 PDF pp. 18–19 · Practice Test 2

To find a cutoff from a percentage, work backwards. Convert the percentage to an **area to the LEFT**, find that area inside the z-table, read off z, and convert back:

$$x = \mu + (z)(\sigma) \qquad \mu = x - (z)(\sigma)$$

WORDING	AREA TO FIND IN THE TABLE
Bottom 20% / 20th percentile	.20
Top 10% / longest-lived 10%	1 − .10 = .90
67.72% are larger than x	1 − .6772 = .3228
x is larger than 86.65% of values	.8665
Central 95%, symmetric	.0250 and .9750

Course examples

- **Goldfish, top 10%:** area .90 → z ≈ 1.28. x = 60 + (1.28)(20) = **85.6 days**.
- **Goldfish, bottom 20%:** area .20 → z ≈ −.84. x = 60 + (−.84)(20) = **43.2 days**.
- **Goldfish, central 95%:** z = ±1.96, so 60 ± 1.96(20) = 60 ± 39.20 → **20.80 to 99.20 days**.
- **Unknown mean:** x = 449, σ = 36, and 86.65% of values are below x. Area .8665 → z = 1.11. μ = 449 − (1.11)(36) = **409.04**.
- **Unknown x:** μ = 3.58, σ = 1.04, and 67.72% are larger than x. LEFT area .3228 → z = −.46. x = 3.58 + (−.46)(1.04) = **3.1016** (3.10).

> **Self-check.** "Top 10%" means you look up .90, because the table gives the area *below* the cutoff. The central 95% of *individual* lifetimes uses σ = 20. A 95% CI for a mean uses a standard error instead. Both use 1.96, but they answer different questions.

Chapter 7 · Sampling distribution of $\bar{x}$

Source: DS Chapter 7 slides 8–29 · Chapter 7 PDF pp. 8–13 · Mean Sample Problems handout · Practice Test 2 credit-card problem

When asked for "the sampling distribution," state all three

- **Center:** $E(\bar{x}) = \mu_{\bar{X}} = \mu$. $\bar{x}$ is an unbiased estimator of μ.
- **Spread:** $\sigma_{\bar{X}} = \dfrac{\sigma}{\sqrt{n}}$, the standard deviation of sample means over repeated samples of size n.
- **Shape:** normal for any n if the population is normal. Otherwise approximately normal when **n ≥ 30** (Central Limit Theorem). The population itself does not become normal.

Sampling error is the natural difference between $\bar{x}$ and μ from one random sample to the next. It does not mean someone made a mistake.

Course example: credit-card balances

μ = $2,864, σ = $775, n = 50. Because n ≥ 30, $\bar{x}$ is approximately normal with center $2,864 and $\sigma_{\bar{X}} = \dfrac{775}{\sqrt{50}} = 109.6016$.

- **P($\bar{x}$ > $3,000):** $z_{\bar{x}} = \dfrac{3000 - 2864}{109.6016} = 1.24$. LEFT area = .8925, so 1 − .8925 = **.1075**.
- **P(within $100 of μ):** $2,764 ≤ $\bar{x}$ ≤ $2,964 → z = ±.91. Then .8186 − .1814 = **.6372**.
- **Same question with n = 100:** $\sigma_{\bar{X}}$ = 775/10 = 77.5 → z = ±1.29. Then .9015 − .0985 = **.8030**.

What changes when n changes

A larger n leaves the center at μ and shrinks $\sigma_{\bar{X}}$, so more sample means fall close to μ. Multiplying n by 4 cuts the SE in half. The population σ does not change.

> **Common mistake.** Don't divide by 775 in the sample-mean z-score. That describes individual balances, not the mean of 50 balances. The finite-population correction factor appears in the Chapter 7 PDF, but it is not on the formula sheet and the practice handouts leave it out.

Chapter 7 · Sampling distribution of $\bar{p}$

Source: DS Chapter 7 slides 39–44 · Chapter 7 PDF pp. 14–16 · Proportion Sample Problems handout · Practice Test 2 internet problem

p is the population proportion and $\bar{p} = \dfrac{x}{n}$ is the sample proportion, where x is the number of items with the characteristic. Keep proportions as decimals: 66% = .66.

- **Center:** $E(\bar{p}) = p$.
- **Spread:** $\sigma_{\bar{p}} = \sqrt{\dfrac{p(1-p)}{n}}$
- **Shape:** approximately normal when $\mathbf{np \geq 5,\ and\ n(1-p) \geq 5}$. The Chapter 7 slides write "> 5", but follow the formula sheet for a boundary case.
- **Probability:** $z_{\bar{p}} = \dfrac{\bar{p} - p}{\sigma_{\bar{p}}}$, then use the same table rules as in Chapter 6.

Course example: internet use among adults over 70

p = .66, n = 300. Find the probability that $\bar{p}$ is within .04 of p. Round proportions to two decimals and the SE to four.

1. **Shape:** np = 300(.66) = 198 and n(1 − p) = 300(.34) = 102. Both are ≥ 5, so the distribution is approximately normal.
2. **Center:** $E(\bar{p})$ = **.66**
3. **Spread:** $\sigma_{\bar{p}} = \sqrt{\frac{(.66)(.34)}{300}} = \sqrt{.000748}$ = **.0273**
4. **Translate:** "within .04" means .62 ≤ $\bar{p}$ ≤ .70.
5. **Standardize:** z = (.62 − .66)/.0273 = −1.47 and z = (.70 − .66)/.0273 = 1.47, giving areas .0708 and .9292.
6. **Subtract:** .9292 − .0708 = **.8584** (85.84%)

> **Traps.** The n ≥ 30 rule is for means. For proportions, check **both counts**. If the question gives a count cutoff, convert it to $\bar{p}$ = x/n first. Write .04, not 4.

Chapter 8 · Confidence intervals and critical values

Source: DS Chapter 8 slides 9–56 · Chapter 8 PDF pp. 1–8, 15–16 · Sigma Known / Minimum Sample Sizes handout · Practice Test 2

Every interval has the same structure: **point estimate ± margin of error**, where E = (critical value)(standard error). The width of the interval is 2E. A mean CI estimates μ, and a proportion CI estimates p.

For confidence level C, α = 1 − C. Split α evenly between the two tails. The critical probabilities are

$$\frac{\alpha}{2} \quad \& \quad 1 - \frac{\alpha}{2}$$

Look up $1 - \frac{\alpha}{2}$ inside the z-table to get $Z_{\alpha/2}$.

CONFIDENCE	A	A/2	1 − A/2	$Z_{\alpha/2}$
80%	.20	.1000	.9000	**1.28**
90%	.10	.0500	.9500	**1.645**
95%	.05	.0250	.9750	**1.96**

CONFIDENCE	A	A/2	1 − A/2	
98%	.02	.0100	.9900	**2.33**
99%	.01	.0050	.9950	**2.576**

When a question asks for both cutoffs, give them as ±z. Use the positive value to compute E.

σ known: the Verizon problem

$\bar{x}$ = 153 monthly minutes per residential user, σ = 46, n = 85, 95% confidence.

$$\mu = \bar{x} \pm (Z_{\alpha/2}) \left(\frac{\sigma}{\sqrt{n}}\right) = 153 \pm (1.96) \left(\frac{46}{\sqrt{85}}\right) = 153 \pm (1.96)(4.9894) = 153 \pm 9.7792$$

CI = **(143.22, 162.78) minutes**. *Interpretation:* We are 95% confident that the population mean monthly minutes per residential user is between 143.22 and 162.78.

Minimum n for a mean: bread expenditure

90% confidence, σ = $4, E = $1.

$$\mathbf{n} = \left(\frac{(z_{\alpha/2})\sigma}{E}\right)^2 = \left(\frac{(1.645)(4)}{1}\right)^2 = 43.2964$$

Round **up** to **44 families**. If σ is unknown at the planning stage, the slides use a pilot study or range/4 as an estimate.

> **What "95% confident" means.** If you repeated the sampling many times, about 95% of the intervals built this way would contain the fixed population parameter. The interval estimates μ. It is not the range that holds 95% of individual values.

Chapter 8 · Mean CI when σ is unknown (t)

Source: DS Chapter 8 slides 57–67 · Chapter 8 PDF pp. 8–12 · Sigma Unknown t handout · Practice Test 2

$$S_{\bar{X}} = \frac{s}{\sqrt{n}} \qquad \mu = \bar{x} \pm (t_{a/2}) \left(\frac{s}{\sqrt{n}}\right), \quad df = n - 1$$

For a small sample, look for the statement that the population is normal. The course allows the large-sample approach when the population isn't normal.

Reading the t-table

Go to the **df row** and the **upper-tail area α/2** column. The number where they meet is t. The body of the t-table holds critical values, while the body of the z-table holds probabilities.

- Upper-tail area .025, df = 12 → `t = 2.179`
- Lower-tail area .05, n = 51 (df = 50) → `t = −1.676`
- Central 90%, n = 26 (df = 25) → `−1.708 and +1.708`

Aerospace managers: the full course example

The population is normal. n = 18, $\bar{x}$ = 13.56 overtime hours/week, s = 7.8, 90% confidence. σ is unknown → t, α = .10, α/2 = .05, df = 17 → t = 1.740.

$$\mu = 13.56 \pm (1.74)\left(\frac{7.8}{\sqrt{18}}\right) = 13.56 \pm (1.74)(1.83848) = 13.56 \pm 3.19895$$

CI = **(10.36, 16.76) hours**. We are 90% confident that the population mean weekly overtime for aerospace managers is between 10.36 and 16.76 hours.

Large samples still use t

- $\bar{x}$ = 25, s = 3.5, n = 60, 95%: df = 59, t = 2.001, SE = .45185, E = .90415 → **(24.096, 25.904)**
- $\bar{x}$ = 3.419, s = .974, n = 32, 90%: df = 31, t = 1.696, E = .29202 → **(3.127, 3.711)**

t compared with z: t is symmetric and bell-shaped but has heavier tails. At the same confidence level its critical value is larger, so the interval is wider. As df grows, t approaches z.

Chapter 8 · Proportion CI and minimum sample size

Source: DS Chapter 8 slides 68–80 · Chapter 8 PDF pp. 13–18 · Proportions / Minimum Sample Size handout · Practice Test 2

$$p = \bar{p} \pm \left(Z_{\alpha/2}\right)\sqrt{\frac{\bar{p}\,(1-\bar{p})}{\mathrm{n}}}$$

Find $\bar{p} = \dfrac{x}{n}$ if it isn't given. Check that n$\bar{p}$ and n(1 − $\bar{p}$) are both at least 5.

Temporary-hire companies (Practice Test 2)

$\bar{p}$ = .39, n = 87, 95% confidence. Counts: 87(.39) = 33.93 and 87(.61) = 53.07, both above 5.

$$p = .39 \pm (1.96)\sqrt{\frac{(.39)(.61)}{87}} = .39 \pm (1.96)(.0523) = .39 \pm .1025$$

CI ≈ **(.2875, .4925)**, or (.29, .49) to two decimals. We are 95% confident that the population proportion of companies using temporary hires during the holiday season is between about 28.75% and 49.25%.

Minimum n for a proportion

$$n = \frac{\left(z_{\alpha/2}\right)^2 p^*\left(1 - p^*\right)}{E^2}$$

Use a pilot p* if one is given. Otherwise use **p* = .50**, because .50(.50) = .25 is the largest possible p*(1 − p*) and gives the most conservative n.

- **Acme, with pilot:** 98% → $z_{\alpha/2}$ = 2.33, E = .03, p* = .40. n = (2.33)²(.40)(.60)/(.03)² = 1.302936/.0009 = 1447.71 → **1,448**
- **Acme, no pilot:** n = (2.33)²(.50)(.50)/(.03)² = 1.357225/.0009 = 1508.03 → **1,509**. The key prints 1,508 because it rounded too early, and the round-up rule gives 1,509.

How the pieces move

CHANGE	EFFECT
Higher confidence	Larger z or t, larger E, wider interval
Larger n	Smaller SE, smaller E, narrower interval
Smaller target E	Larger required n. Halving E needs 4× the n
Larger σ	Larger required n for a mean

> **Units and rounding.** "Within 3 percentage points" means E = .03, not 3. A minimum sample size always rounds **up** to the next whole number.

Concepts and vocabulary

Source: Chapter 7 PDF pp. 1–8 · Chapter 7–8 slides · Chapter 6 PDF pp. 1–2, 20

TERM	MEANING
Population / sample	All elements of interest / the subset studied
Parameter	Describes the population: μ, σ, p
Statistic	Describes the sample: $\bar{x}$, s, $\bar{p}$
Sampling distribution	Distribution of a statistic over all possible samples of the same size
Standard error	Standard deviation of a sampling distribution
Point / interval estimate	A single statistic / a range built around it
Unbiased estimator	Its expected value equals the parameter. It doesn't have to hit the parameter in every sample

Sampling methods

METHOD	HOW TO RECOGNIZE IT
Simple random	Every item in the frame has an equal chance of selection
Stratified random	Divide into non-overlapping subgroups (strata) and randomly sample from *each* one
Systematic	Random start, then every kth item, where k = N/n
Cluster	Randomly select *some* groups or areas that represent the population
Convenience	Whoever or whatever is easiest to reach
Judgment	The researcher picks people judged to be representative
Quota	Fill subgroup targets using nonrandom selection

Stratified vs. cluster: stratified sampling draws from every stratum, while cluster sampling selects only some clusters. **Stratified vs. quota:** stratified selection within groups is random, and quota selection is not.

Four survey errors

- **Coverage/selection:** the frame leaves out relevant groups.

- **Nonresponse:** people who don't answer differ from those who do.
- **Measurement:** question wording, respondent mistakes, or interviewer effects distort the answers.
- **Sampling:** estimates naturally vary from sample to sample.

Reasons to sample include saving time and money, and avoiding the loss of product in destructive testing.

Standard deviation from raw data

Compute the mean, each deviation, each squared deviation, and their sum. Then divide by **N** for a population or **n − 1** for a sample, and take the square root (see sheet page 1). A normal probability plot whose points lie close to the reference line supports approximate normality.

Practice Test 2: method map

Source: Practice Test 2 V3 Revised · Answer Key V3 Revised 2

Work Practice Test 2 on blank paper using only the formula sheet and the tables. Name the method before you calculate. Then open each item below to check your answer. The practice test says it shows some kinds of exam questions, not everything that may be asked.

PROBLEM	WRITE FIRST
Bread expenditure	Mean sample size, 90%, $Z_{\alpha/2}$ = 1.645
Aerospace managers	Mean CI, s given → t, df = 17
x = 449, 86.65% below	Inverse normal, solve for μ
$\bar{x}$ = 25, s = 3.5, n = 60	t, df = 59, 95%
$\bar{x}$ = 119, σ = 23.89, n = 75	z, 98%
$\bar{x}$ = 3.419, s = .974, n = 32	t, df = 31, 90%
$\bar{x}$ = 57.7, σ = 12.1, n = 47	z, 80%
Temporary-hire companies	Proportion CI, $\bar{p}$ = .39
67.72% larger than x	LEFT area .3228, inverse normal
Goldfish lifetimes	Individual x, σ = 20
Acme sample size	Proportion n, E = .03
Credit-card sample means	$\sigma_{\bar{X}} = \sigma/\sqrt{n}$

PROBLEM	WRITE FIRST
Verizon	Mean CI with z
Internet-use sample proportion	Center, SE, shape, then probability

▼ **Answers: Chapter 6 problems**

Goldfish: .50 · .1587 · .1865 · .0040 = .40% · 20.80–99.20 days · 85.6 days · 43.2 days

x = 449 → μ = 409.04 · 67.72% larger → x = 3.1016

▼ **Answers: Chapter 7 problems**

Credit card: .1075 · .6372 · .8030 (n = 100)

Internet use: center .66 · SE .0273 · approximately normal · .8584

▼ **Answers: Chapter 8 problems**

Bread: n = 44 · Aerospace: (10.36, 16.76) · Verizon: (143.22, 162.78)

(24.096, 25.904) · (112.57, 125.43) · (3.127, 3.711) · (55.44, 59.96)

Temporary hires: (.2875, .4925) · Acme: 1,448 with pilot, 1,509 without

Fix weak spots with the matching Canvas handout

- Normal areas and cutoffs → *Chapter 6 Normal Curve Examples with Answers*
- Sample-mean probabilities → *Chapter 7 Mean Sample Problems with Answers*
- Sample-proportion probabilities → *Chapter 7 Proportion Sample Problems with Answers*
- z CI and minimum n for a mean → *Chapter 8 Mean Sigma Known / Minimum Sample Sizes*
- t CI → *Chapter 8 Mean Sigma Unknown t*
- Proportion CI and minimum n → *Chapter 8 Proportions / Minimum Sample Size*

Final checks

Source: Exam 2 announcement · Practice Test 2 key · Chapter 6 PDF p. 19

Errors in the practice key not to learn from

- **Acme, no pilot:** the correct answer is 1,509, not 1,508. Always round n up at the very end.

- **67.72% above x:** one line of the key prints .3328. The correct LEFT area is **.3228**, which gives z = −.46.
- **Goldfish above 60:** the answer .50 is right, but the key shows the z-score as .5. The correct value is **z = 0**.
- **Verizon:** the interval estimates *mean monthly minutes per residential user*, not the average length of a call.

Rounding

Professor Spencer distinguishes **"round final answer only"** (keep full precision until the end) from **"round all calculations"** (round at every step). Follow the exact number of decimals each question asks for. Canvas may reject an answer that is rounded differently. Minimum sample sizes always round **up**.

Before class

- [ ] LockDown Browser installed and tech check done. Laptop charged.
- [ ] Calculator, blank paper, and a pen or pencil.
- [ ] Clean z-table, t-table, and formula sheet with no notes on them.
- [ ] Arrive on time. Once logged in, turn your phone off and put it in your bag.

During the 95 minutes

- For each problem, write the formula from the sheet before you touch the calculator.
- Sketch the curve and shade the area for every probability.
- Number your work, put your name on every page, and hand it in.

Canvas sources

`MGSC2301 Merged Fall 2026 · links need your Canvas login`

All content comes from the course's Canvas materials as gathered in the uploaded study-guide PDF. No outside sites or textbooks were used.

- Professor Spencer: Exam 2 Preparation (announcement, Oct 28, 2026)
- Daily Schedule of Covered Material, MGSC2301 Fall 2026
- Formula Sheet for Test 2 (3 pages)
- Practice Test 2, V3 Revised · Practice Test 2 Answer Key V3 Revised 2
- DS Chapter 6 Normal Curve 2026 slides · DS Chapter 7 2026 slides · DS Chapter 8 Confidence Intervals 2026 slides

- Chapter 6 PDF: Normal Distribution · Chapter 7 PDF: Sampling and Sampling Distributions · Chapter 8 PDF: Estimation for Single Populations
- Chapter 6 Normal Curve Examples · Chapter 7 Mean / Proportion Sample Problems · Chapter 8 Sigma Known, Sigma Unknown t, and Proportions handouts (all with answers)
- Clean course z-table and t-table
